

ALLEN S. MAO

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SUMMARY

R&D Engineer at Ansys (Central ML Team) with experience developing cloud infrastructure for Machine Learning-powered CAE / multiphysics simulation. Former SDE intern at Amazon and NASA.

EDUCATION

University of California Berkeley

August 2019 - May 2022

BA—Computer Science, GPA: 3.72/4.0

Berkeley, California

- *Undergraduate Researcher* with Space Sciences Lab and International Computer Science Institute
- Elected *International House Resident Councilmember, Political Computer Science Vice President*
- Coursework in Machine Learning, Robotics, Embedded Systems, Probability, Databases, Security

WORK EXPERIENCE

Synopsys (formerly Ansys)

Sep 2022 – Present

Sr. AI Engineer, SimAI Team

Paris, France (remote)

- Lead developer of data preprocessing pipeline, converting raw multiphysics solver data into machine learnable data.
- Optimized large-scale mesh processing using **VTK**, identifying and reporting upstream bugs to **Kitware** to improve library stability for the engineering community.
- Engineered data transformation workflows that bridge the gap between traditional FEA/CFD solvers and Graph Neural Network (GNN) based architectures.

R&D Engineer II, Office of the CTO

[Location]

- Architected and deployed a cloud-native web platform on **AWS ECS** to democratize access to internal multiphysics neural-network solvers.
- Integrated ML models into production environments using **Amazon SageMaker**, enabling real-time inference and scalable model hosting for R&D researchers.
- Collaborated with research scientists to translate complex Python-based solvers into a user-facing web application with robust backend architecture.

Amazon, Inc.

May-August 2021

Software Development Engineering Intern

Portland, Oregon (virtual)

- Conducted comprehensive user research by interviewing internal customers to collect user stories and feedback aimed at enhancing the vendor search experience.
- Designed, reviewed, and implemented a smart search system using a Vue.js and Spring stack, resulting in a more efficient and user-friendly vendor search interface

NASA John F. Kennedy Space Center

August-December 2020

Software Engineering Intern

Merritt Island, Florida (virtual)

- Developed and executed comprehensive system and acceptance tests for the graphical user interface of the Launch Control System in the *Artemis I* lunar mission's Space Launch System.

University of Southern California, Information Sciences Institute

June-August 2019

NLP Research Intern (NSF REU)

Marina del Rey, California

- Developed **SoMEF** (Software Metadata Extraction Framework), an automated pipeline to extract description, installation, invocation, and citation metadata from scientific software documentation[cite: 1, 7, 26].
- **First-authored** and presented "SoMEF: A Framework for Capturing Scientific Software Metadata from its Documentation" at the **2019 IEEE International Conference on Big Data**[cite: 1, 2, 240].

ADDITIONAL ACTIVITIES AND MISCELLANEOUS SKILLS

- **Yellowstone NP** Intern Ranger, staff for the Youth Conservation Corps (May-Aug 2022). Blog: [↗](#)
- **Publications:** "Software Metadata Extraction Framework" [↗](#); "COVID-19's Long Term Impacts on Economic Inequality in the United States": Essay Contest Winner in the *Berkeley Economic Review*; NASA Technical Report #20205009014 [↗](#)
- **Projects:**  *Autonomous Robot Platoon Merges*: Embedded Systems Course Project; *CovidAccountable*: Chrome extension that identifies PPP fund recipients (**Global Hacks Overall Best Prize**)